

VedaAI — AI Assessment Extraction Test

Computer Science Fundamentals • Practice Assessment

Time: 60 minutes **Maximum Marks:** 40

Instructions

1. Answer the questions in your own words. 2. You may answer questions in any order. 3. Label each answer clearly. 4. For multi-part questions, answer each labelled part separately.

1. What is the primary purpose of an operating system? **[2 marks]**
2. Explain the difference between a process and a thread. **[3 marks]**
- 3 (a). What is normalization in a relational database? **[2 marks]**
- 3 (b). Give one example of a situation where denormalization can be useful. **[2 marks]**
4. Write the time complexity of binary search and briefly explain why. **[3 marks]**
5. What is an API? Give one practical example. **[3 marks]**
- 6 (a). Differentiate supervised learning and unsupervised learning. **[3 marks]**
- 6 (b). Name one supervised learning algorithm and one unsupervised learning algorithm. **[2 marks]**
7. Why are indexes used in databases? Mention one trade-off of using indexes. **[4 marks]**
8. What is overfitting in machine learning, and mention one way to reduce it? **[3 marks]**
9. Explain what a REST API means by the term *stateless*. **[3 marks]**
10. Consider the array [2, 4, 6, 8, 10]. What is its mean and median? **[3 marks]**
- 11 (a). What is a hash table? **[2 marks]**
- 11 (b). What happens when two keys produce the same hash value? **[2 marks]**
12. Name two common HTTP methods and describe what each is generally used for. **[3 marks]**